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# Understanding K-Food Semantics in Resource-Constrained Environment

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## Abstract

The global spread of K-Content has increased tourism in Korea, yet visitors often face a significant culinary barrier due to language and a lack of contextual information about Hansik (Korean food). Addressing this challenge requires an intelligent system capable of operating in offline, resource-constrained environments, such as a mobile device. We propose a hybrid **Classification-Retrieval-Generation (CRG)** pipeline designed to provide accurate and fluent food descriptions from an image. This pipeline first uses a visual classifier to identify the dish, then retrieves fact-grounded information from a curated knowledge base, and finally employs a small language model (TinyLlama-1.1B) to synthesize this information into a natural-language explanation. We conducted a comparative analysis of classifier architectures on the AI-Hub Korean Food Image dataset, identifying the Finetuned CLIP as the optimal model with 86.63% test accuracy. To address the model’s closed-set limitation, we further investigated the WiSE-FT (Weight-Space Ensembling) architecture. However, while the model retained high performance on seen examples, our experiments revealed that it remained insufficient for generalizing to out-of-distribution (OOD) dishes. Consequently, to overcome this limitation and achieve robust open-set recognition on mobile devices, we propose a VLM distillation pipeline as a key direction for future work.

## 1 Introduction

### 1.1 Background and Problem Definition

The global spread of K-Content has led to a rapid increase in foreign visitors to Korea. However, many tourists face a culinary barrier when encountering Hansik (Korean food), primarily due to insufficient information on dish names and ingredients, coupled with language barriers. This issue is difficult to solve with simple translation or basic image recognition alone, necessitating an intelligent approach that incorporates the dish’s cultural context.

Our project aims to resolve this Hansik comprehension barrier by developing an AI system that provides accurate and natural descriptions upon receiving a dish’s image. A primary requirement is that this system must operate on a tourist’s mobile device, enabling real-time inference in an offline environment where internet connectivity may be unavailable. This necessitates a key focus on designing an efficient architecture capable of operating within a resource-constrained environment, without relying on massive, server-dependent models. To achieve this, we adopt a **Classifica-**

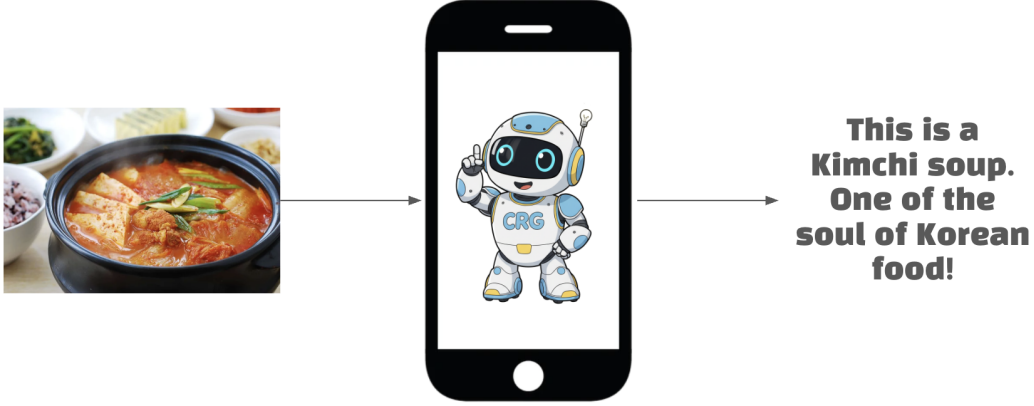


Figure 1: High-level description of our **CRG (Classification-Retrieval-Generation)** pipeline. It is devised for the resource constrained environment such as a mobile phone.

**Classification-Retrieval-Generation (CRG) pipeline:** using the classification result to retrieve information from a Knowledge Base (DB), and leveraging a small Language Model to generate factual and fluent explanations.

## 1.2 Proposed Approach: The Hybrid CRG Pipeline

To address the inherent limitations of standard text-only LLMs, namely their inability to understand image inputs, high computational cost, and the hallucination problem, we propose and implement a **Hybrid Classification-Retrieval-Generation (CRG) pipeline**.

The pipeline is structured as follows:

1. **Classification:** A lightweight classifier quickly and accurately identifies the food item from the input image.
2. **Retrieval:** Using the identified food name as a key, the system retrieves fact-grounded descriptions from a carefully curated Knowledge Base (DB).
3. **Generation:** A small Language Model (TinyLlama/TinyLlama-1.1B-Chat-v1.0) is fed the retrieved factual information to generate a fluent, user-friendly, and contextual final response.

## 1.3 Key Contributions and Report Structure

This report details the preliminary results from the construction and optimization of the CRG pipeline. Our main contributions include:

1. **Optimal Classifier Selection:** A comparative analysis of classifiers (CNN, ViT, and CLIP) to select the most efficient model that provides optimal performance under resource constraints.
2. **Implementation of WiSE-FT for Robustness:** To address the trade-off between domain-specific accuracy and the loss of general knowledge (catastrophic forgetting), we adopt the WiSE-FT [8] methodology. We explore how interpolating the weights of a fine-tuned model with its zero-shot counterpart affects the system’s ability to handle both seen and unseen data distributions.
3. **Qualitative Results for CRG pipeline:** We present a detailed analysis of generated text for sample images. This analysis will evaluate the fluency, factual accuracy, and contextual relevance of the outputs, demonstrating the effectiveness of the hybrid CRG pipeline.
4. **Failure Analysis and Proposal for VLM Distillation:** We critically analyze the limitations of current closed-set and WiSE-FT approaches, particularly regarding their performance on out-of-distribution (OOD) queries. Based on these insights, we propose a VLM Distillation pipeline as a future direction to transfer the open-set generalization capabilities of large Vision-Language Models into our lightweight framework.

## 2 Related works

Our research aims to develop an efficient and fact-grounded system for semantic K-Food understanding. This goal sits at the intersection of key studies in Fine-grained Classification and the Retrieval-Augmented Generation (RAG) paradigm.

### 2.1 Food Classification and Domain-Specific Datasets

Initial developments in food recognition established benchmarks like the Food-101 dataset [2], driving the progress of CNN-based classification models. While these classification approaches achieve high accuracy for seen classes, they often struggle with the fine-grained differentiation required for visually similar dishes in specialized domains like Hansik. Furthermore, they fundamentally lack the ability to generate descriptive text for classes outside their training distribution, highlighting the limitation of a purely discriminative approach.

### 2.2 Retrieval-Augmented Generation (RAG) and Hybrid Approaches

To mitigate the hallucination problem and enhance the factuality of generated text, the **Retrieval-Augmented Generation (RAG)** paradigm [5] was introduced. RAG ensures that the generation process is grounded in information retrieved from a trusted, external knowledge store. Our Classifier-DB-LLM (CRG) Hybrid Pipeline adapts this philosophy to the vision domain. By decomposing the task into a lightweight Classification step, a Retrieval step from a curated database, and a small Generation step, our approach seeks to achieve an optimal tradeoff between the efficiency of small models and the factual reliability provided by external knowledge.

### 2.3 Vision-Language Models and Zero-Shot Transfer

Traditional supervised learning models, as discussed in Section 2.1, are fundamentally limited by a fixed set of predefined categories. To overcome this closed-set constraint, recent advancements have focused on Vision-Language Models (VLMs) that learn joint representations of images and text. A prominent example is **CLIP (Contrastive Language-Image Pre-training)** [6], which is trained on a massive dataset of 400 million image-text pairs.

Unlike standard classifiers that map images to discrete label indices, CLIP utilizes a contrastive loss objective to align visual and textual embeddings in a shared feature space. This architecture enables "zero-shot" transfer, where the model can classify images into categories it has never seen during training by matching the image embedding with the most similar text prompt (e.g., "A photo of dish name"). In the context of semantic K-Food understanding, incorporating CLIP offers a potential solution to the out-of-distribution (OOD) problem, allowing the system to recognize and describe dishes beyond the initial training dataset without requiring task-specific fine-tuning.

## 3 CRG Pipeline

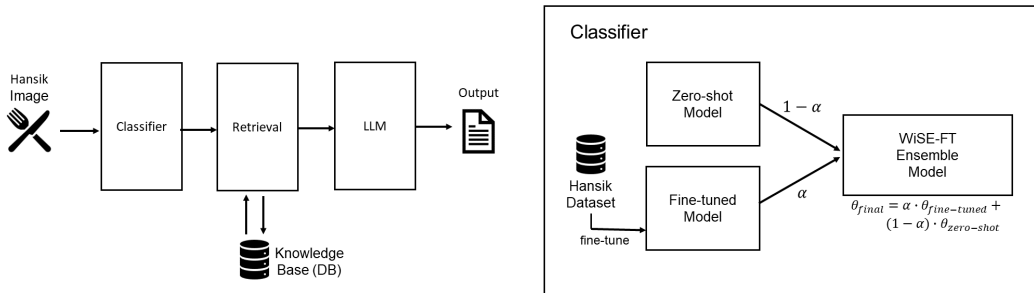


Figure 2: Overview of the proposed framework. Left: CRG pipeline for Hansik images. Right: WiSE-FT ensemble method used to construct the classifier component, combining zero-shot and fine-tuned models.

The CRG pipeline operates in three sequential, specialized stages, as illustrated in Figure 2.

### 3.1 Stage 1: Classification

The classification stage serves as the entry point of our pipeline, processing the input image to determine the specific dish category. To identify the most suitable architecture for semantic K-Food understanding in a resource-constrained environment, we investigated two distinct classification paradigms: traditional Supervised Classification and Zero-Shot Classification.

#### 3.1.1 Approach: Supervised Classification Architectures

For the closed-set classification task, we evaluated three distinct architectural families to analyze the accuracies.

- **ResNet:** We selected ResNet[4] as a standard CNN baseline. This architecture utilizes residual connections to enable deep feature extraction. We included it to establish a performance benchmark for standard convolutional operations against newer architectures.
- **EfficientNet:** Targeting the requirement for on-device inference, we employed the EfficientNet[7] family. This architecture utilizes a compound scaling method that uniformly scales network width, depth, and resolution. It is designed to maximize parameter efficiency, making it a critical candidate for our mobile-oriented goals.
- **ViT:** To investigate the efficacy of non-convolutional architectures, we incorporated the Vision Transformer (ViT)[3]. Unlike CNNs, ViT processes images as sequences of patches using self-attention mechanisms. We structured our experiment to include variants of different capacities (Base, Small, and Tiny) to analyze how model size correlates with feature representation in fine-grained food datasets.

#### 3.1.2 Approach 2: Robust Open-Set Classification via WiSE-FT

To enhance the model’s scalability beyond the closed-set limitation, we utilized the CLIP [6] framework. However, standard fine-tuning on a specific domain risks “catastrophic forgetting,” where the model loses its pre-trained ability to recognize out-of-distribution (OOD) dishes (e.g., *Dakbal*) that are not in the training set. To resolve this trade-off, we adopted the **WiSE-FT (Weight-Space Ensembling)** [8] architecture.

- **Methodology:** Rather than relying solely on a fine-tuned model, we construct a robust classifier by linearly interpolating the weights of two distinct model states: the original pre-trained zero-shot model ( $\theta_{\text{zero-shot}}$ ) and the model fine-tuned on the AI-Hub Korean Food dataset ( $\theta_{\text{fine-tuned}}$ ).
- **Architecture & Mechanism:** The final model weights ( $\theta_{\text{final}}$ ) are computed directly in the weight space using a mixing coefficient hyperparameter  $\alpha$  (where  $0 \leq \alpha \leq 1$ ), according to the following formulation:

$$\theta_{\text{final}} = \alpha \cdot \theta_{\text{fine-tuned}} + (1 - \alpha) \cdot \theta_{\text{zero-shot}}$$

- **Rationale:** By ensembling in the weight space, this architecture combines the strengths of both models. It retains the broad, open-set generalization capability of the original CLIP (essential for recognizing unseen dishes) while integrating the high discrimination accuracy for specific K-Food classes acquired through supervised fine-tuning.

### 3.2 Stage 2: Retrieval

The identified dish name from the Classification stage acts as the key for the Retrieval module. This module accesses a carefully curated **Knowledge Base (DB)**, which stores structured, fact-grounded information about each dish. This knowledge base is implemented as a simple Key-Value mapping, where the dish name is the key. The stored data follows a rich structure, including fields such as `korean_name`, `english_name`, `description`, `category`, `ingredients`, `cooking_method`, and `cultural_note`. The retrieved factual information is then formatted as a structured prompt (e.g., a JSON or structured text block) to be passed to the generative model. This explicit retrieval step is crucial for guaranteeing high factual accuracy and mitigating the hallucination inherent in large language models.

### 3.3 Stage 3: Generation

In the final stage, a small Language Model (LLM) is used to generate the final, fluent explanation. We utilize a TinyLlama/TinyLlama-1.1B-Chat-v1.0 as our generator. By supplying the LLM with the retrieved factual data, the resulting generation is augmented: its task shifts from generating potentially ungrounded knowledge to merely fluently articulating the provided facts. This addition of ground truth data is crucial for achieving high description quality and factual richness while adhering to our resource constraints.

## 4 Experiments

### 4.1 Dataset

We utilize the AI-Hub Korean Food Image Dataset [1] for our experiments. Designed for fine-grained visual classification, this dataset comprises 150,000 images evenly distributed across 150 distinct categories of Korean cuisine, with 1,000 images per class.

Since the original dataset is provided in Korean, we first translated all class labels into English. Furthermore, a critical limitation of the original dataset for our purpose is that it contains only image-label pairs, lacking the descriptive paragraphs required for our image-to-text generation task. To bridge this gap, we augmented the dataset by generating detailed English descriptions for each food category using OpenAI’s GPT-5. These synthesized texts serve as the knowledge base and ground truth for our CRG pipeline.

### 4.2 Determining optimal classifier

| Model Name            | Train Acc (%) | Test Acc (%) |
|-----------------------|---------------|--------------|
| ResNet50              | 99.67         | 84.78        |
| EfficientNetB0        | 98.85         | 85.96        |
| EfficientNet B1       | 98.47         | 85.44        |
| EfficientNet B2       | 99.20         | 86.32        |
| EfficientNet B3       | 99.32         | 86.36        |
| EfficientNet B4       | 97.66         | 85.08        |
| EfficientNet B5       | 99.39         | 85.12        |
| ViT base              | 99.70         | 77.60        |
| ViT Small             | 81.57         | 84.11        |
| ViT Tiny              | 74.79         | 79.17        |
| <b>Finetuned CLIP</b> | 96.90         | <b>86.63</b> |

Table 1: Classifier performance for various baseline models

We finetuned a baseline classifiers including ResNet, EfficientNet, ViT and CLIP using AI-Hub Korean Food Image Dataset [1]. The CLIP fine-tuning approach employs contrastive learning to align visual and textual representations by computing cosine similarity scores between image features extracted via the vision encoder and pre-computed text features for food class names, optimized using cross-entropy loss with temperature scaling to maximize alignment between matching image-text pairs. The results are summarized in Table 4.2. The finetuned CLIP achieved the highest test accuracy at 86.63%. Therefore, we selected the finetuned CLIP as the best-performing model for our task.

### 4.3 CLIP-based classifier zero-shot performance

Even though the finetuned CLIP turned out to be the best performing model within the dataset, its zero-shot ability is decreased. To mitigate this problem, we adopted the WiSE-FT method. To evaluate the zero-shot performance, we built the out-of-distribution image dataset with 10 classes and total size 50.

The results presented in Table 4.3 illustrate the tension between task-specific optimization and zero-shot robustness. We observe that the fine-tuning process resulted in significant catastrophic forgetting:

| Alpha                        | Top-1 Acc (%) | Top-5 Acc (%) |
|------------------------------|---------------|---------------|
| $\alpha = 0$ (No finetune)   | <b>36</b>     | <b>38</b>     |
| $\alpha = 0.3$               | 22            | 36            |
| $\alpha = 0.5$               | 8             | 32            |
| $\alpha = 0.7$               | 8             | 24            |
| $\alpha = 1$ (Full finetune) | 4             | 18            |

Table 2: Zero-shot performance of finetuned CLIP with WiSE-FT.

the fully fine-tuned model ( $\alpha = 1$ ), despite achieving 86.63% accuracy on the target dataset, exhibits a near-total collapse in zero-shot capability with a Top-1 accuracy of only 4%.

Although pure pre-trained weights ( $\alpha = 0$ ) yield the best zero-shot performance (36%), relying solely on them would negate the benefits of the domain adaptation achieved in the previous section. It is worth noting that for configurations where  $\alpha \neq 0$ , the in-distribution validation accuracy is maintained at the high level of the fine-tuned model. Therefore, we selected  $\alpha = 0.3$  as the optimal operating point. This choice represents a strategic trade-off: it mitigates the extreme forgetting observed in the fully fine-tuned model (recovering Top-1 accuracy from 4% to 22%) while still incorporating a significant portion of the fine-tuned weights to maintain high classification accuracy on the learned Korean Food classes.

#### 4.4 Qualitative Results

| Image                                                                               | Generated text                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | Bibimbap is a popular Korean rice dish that is both delicious and visually appealing. It combines various seasonal vegetables, spicy beef, and crispy fried eggs on top of a hot bowl of steaming rice. This dish has become synonymous with Korean cuisine, representing harmony and balance through its vibrant colors and textures.                                    |
|  | As a Korean food expert, I am thrilled to share with you my knowledge about one of Korea’s most beloved dishes - Acorn Jelly. Acorn Jelly is a unique and delicious Korean dish that combines traditional Korean flavors and techniques with modern cooking methods. This dish is often served as an appetizer or side dish at traditional Korean BBQ or banchan parties. |

Table 3: Generated texts from sample images

To assess the practical quality of our Classifier-Retriever-Generator (CRG) model, we present a selection of generated explanations for various Korean food images in Table 4.4. The qualitative analysis demonstrates that the pipeline is highly effective at producing accurate, fluent, and contextually rich descriptions when the initial classification is correct.

As shown in the examples for “Bibimbap” and “Acorn Jelly,” the system consistently executes the retrieval and generation stages with precision. The classifier first correctly identifies the visual input, which allows the retriever to gather specific cultural facts—such as Bibimbap representing “harmony and balance” or Acorn Jelly being served as a “side dish” (banchan).

This successful retrieval provides the necessary factual grounding for the generator. The 1.1B LLM synthesizes these disjointed facts into natural paragraphs. The high quality of these outputs validates the core strength of our CRG architecture: it effectively leverages a lightweight generator to produce text that is both stylistically fluent and factually dense, achieving performance comparable to significantly larger models by offloading knowledge storage to the retrieval module.

## 4.5 Failure Case Analysis

| Image                                                                             | Generated text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | As a Korean food expert, I can provide you with a detailed description of Fish Pancake, one of the most beloved Korean dishes. Fish Pancake is a savory pancake made with various types of seafood, including fish or shrimp, that are pan-fried until they become crispy on the outside. The dish originated from Seoul's Han River area, where fishermen used to catch fresh                                                                                                                             |
|  | "Budae Jjigae" is a popular Korean noodle soup that features tender pork belly and a savory broth made from various vegetables, including garlic, onions, ginger, and scallions. The dish is typically served with rice and often includes a variety of side dishes like boiled egg, pickled radish, and bean sprouts. This dish is similar to the Kimchi Stew, which is another popular Korean stew featuring fermented cabbage. Both dishes are hearty and flavorful, perfect for a cold winter evening. |

Table 4: Failure cases of our CRG pipeline.

The examples in Table 4.5 reveal the primary weakness of our CRG pipeline: its total dependence on the initial classifier’s accuracy and the structural limitations of our retrieval system. We observed two main types of failure:

- **Misclassification when performing zero-shot:** As seen in the first row, the input image contains *Hotteok*. However, the classifier incorrectly identifies it as “Fish Pancake.” Since our Knowledge Base relies on a strict Key-Value structure, this incorrect prediction acts as a valid key, retrieving factual data for the wrong dish. Consequently, the generator produces a fluent and detailed description of a Fish Pancake, which is factually unrelated to the visual input.
- **Extreme hallucination due to Knowledge Base limitations:** The second row (Budae Jjigae) demonstrates the challenge of using a Key-Value based Knowledge Base for zero-shot classes. Because the specific key for this unseen class does not exist in our database, we cannot exploit the Knowledge Base directly. To mitigate this, we attempted to retrieve context by using the Top-3 classification results as proxy keys to include information about similar foods. However, as shown in the description, the hallucination persisted. The model failed to utilize the proxy context effectively and instead fabricated a generic description, omitting the dish’s defining characteristics.

## 5 Discussions

### 5.1 Limitations of Current Architecture

The limitations of our current CRG pipeline for Out-of-Distribution (OOD) data stem from two distinct but critical issues that independently lead to system failure.

First, the classification performance remains insufficient for open-set recognition. Despite applying WiSE-FT, our experiments showed that while Top-5 robustness improved, the Top-1 accuracy for unseen dishes is critically low. The model inevitably forces OOD inputs (e.g., Dakbal) into one of the 150 training categories based on visual similarity, leading to fundamentally incorrect predictions.

Second, there is a structural "retrieval void" caused by the finite Knowledge Base. Even if the classifier were hypothetically capable of correctly identifying an unseen dish (e.g., predicting "Budae Jjigae"), the current architecture would still fail because the Knowledge Base contains fact-grounded information only for the original 150 classes. Consequently, for any unseen dish, there is no data

to retrieve. Deprived of ground truth, the LLM is forced to generate descriptions without factual support, resulting in severe hallucination. These two problems—inaccurate classification and the lack of retrieval data—makes the current pipeline unreliable for open-world exploration.

## 5.2 Future Works

To simultaneously address both the classification limitation and the retrieval void, we suggest a Vision-Language Model (VLM) Distillation pipeline for the future work.

We propose a "Teacher-Student" framework to distill the open-set capabilities of a large VLM like GPT-5.1 (Teacher) into our on-device model (Student). This approach solves the two problems as follows:

1. **Solving Classification Limits:** By learning the Teacher’s soft probability distributions and feature representations, the Student learns to understand visual semantics (e.g., "red," "spicy," "grilled") rather than just memorizing a fixed set of hard labels.
2. **Solving the Retrieval Void:** By internalizing the Teacher’s knowledge directly into the model’s weights, the system bypasses the need for a static text database. The Student can generate descriptions based on the learned visual features of the unseen dish, effectively eliminating the hallucination caused by missing database entries.

## 5.3 Github Repository Link

- Changwoo Yoo: <https://github.com/changwoolab/20252R0136COSE36203>
- Jaeseung Lee: <https://github.com/j-seui/20252R0136COSE36203>
- Jiyeon Lee: <https://github.com/Jyoonly/20252R0136COSE36203>

## 6 References

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